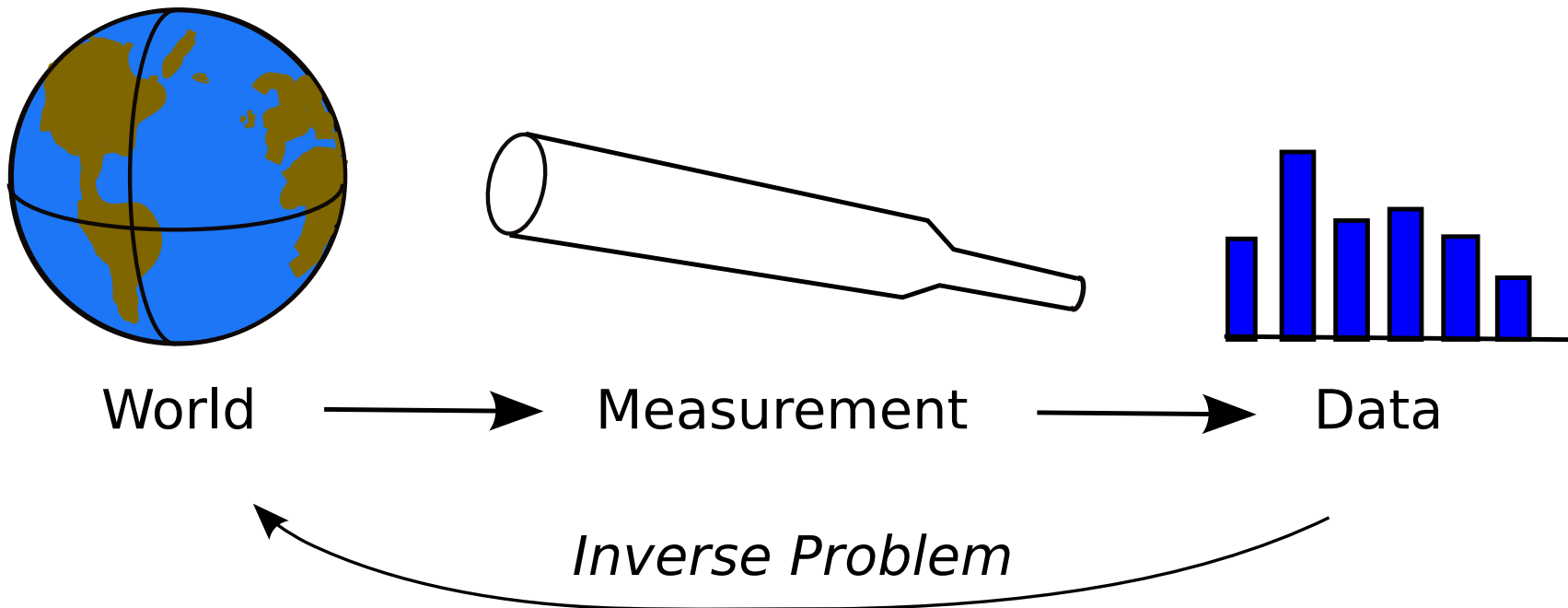


# Mathematics of Machine Learning

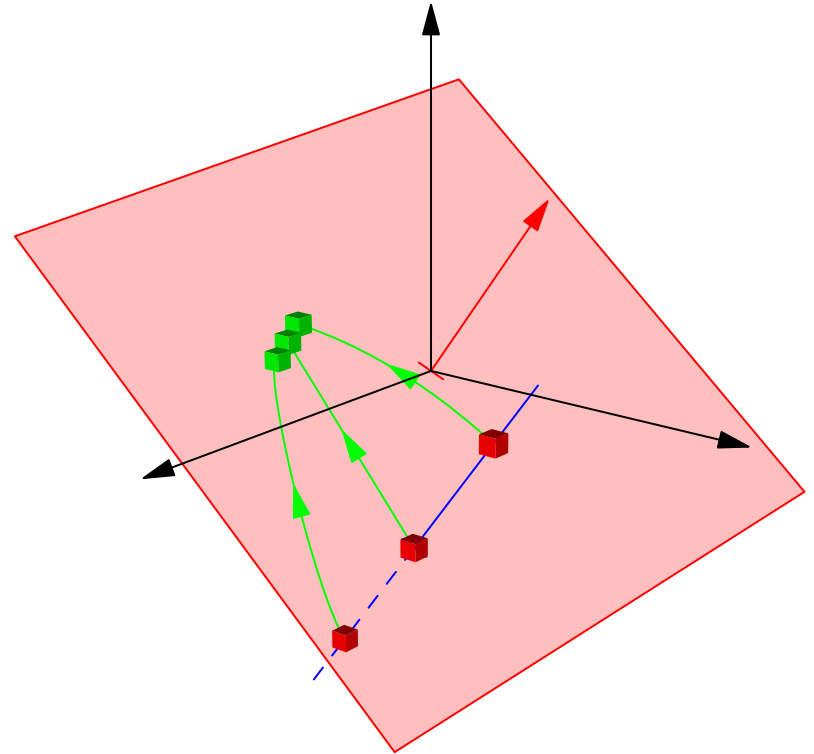
## *Understand Mappings*



*Mappings, Linear Maps, Solving Linear Systems*

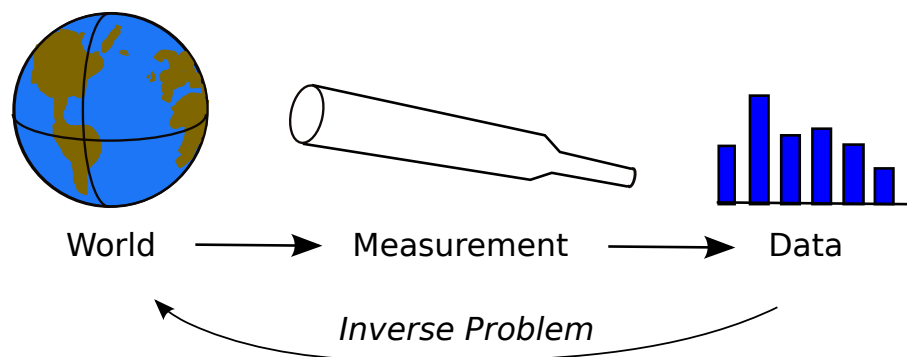
# Outline

1. **Mappings**
2. Linear Maps



# Transforming Data

- In the last lecture we spent time developing a sophisticated view of vector spaces and operators
- At a mathematical level machine learning can be viewed as performing an inverse mapping



- Although our mappings are not necessarily linear in either direction we learn a lot by understanding linear operators

# Inverse Problems

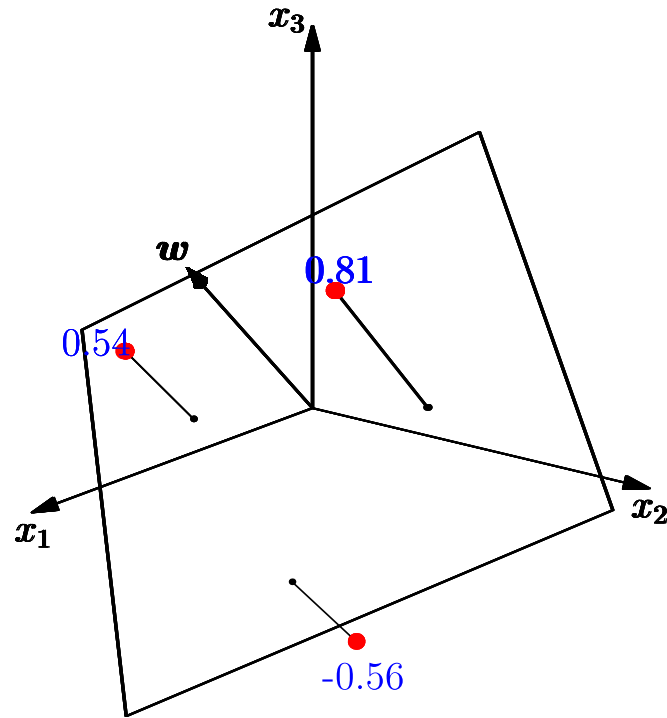
- Given  $m$  observations  $\{(\mathbf{x}_k, y_k) | k = 1, \dots, m\}$  and  $p$  unknown  $\mathbf{w} = (w_1, w_2, \dots, w_p)^\top$  such that  $\mathbf{x}_k^\top \mathbf{w} = y_k$  then to find  $\mathbf{w}$
- Define the *design matrix* as the matrix of feature vectors

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{pmatrix} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mp} \end{pmatrix}$$

- and the target vector  $\mathbf{y} = (y_1, y_2, \dots, y_m)^\top$
- Then if  $m = p$  we have  $\mathbf{y} = \mathbf{X}\mathbf{w}$  or  $\mathbf{w} = \mathbf{X}^{-1}\mathbf{y}$

# Linear Regression

- $x_k^T w$  depends on distance from separating



- If  $m > p$  then  $X$  isn't square so doesn't have an inverse
- Worse unless the data is accurate  $y \approx Xw \Rightarrow$  no "solution"
- Problem solved by Gauss to predict the orbit of the asteroid Ceres

# Linear Least Squares

- The error of input pattern  $\mathbf{x}_k$  is

$$\epsilon_k = \mathbf{x}_k^\top \mathbf{w} - y_k$$

- The squared error

$$E(\mathbf{w}|\mathcal{D}) = \sum_{k=1}^m (\mathbf{x}_k^\top \mathbf{w} - y_k)^2 = \sum_{k=1}^m \epsilon_k^2 = \|\boldsymbol{\epsilon}\|^2$$

- We can define the error vector

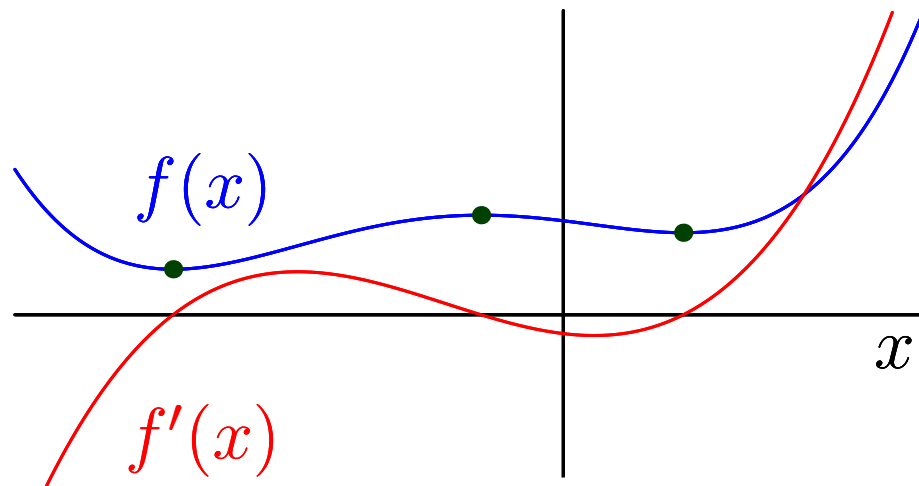
$$\boldsymbol{\epsilon} = \mathbf{X}\mathbf{w} - \mathbf{y}$$

(note that  $\epsilon_k = \mathbf{x}_k^\top \mathbf{w} - y_k$ )

- Minimising this error is known as the least squares problem

# Finding a Minimum

- The minima of a one dimensional function,  $f(x)$ , are given by  $f'(x) = 0$



- The minima of an  $n$ -dimensions function  $f(\mathbf{x})$  are given by the set of equations

$$\frac{\partial f(\mathbf{x})}{\partial x_i} = 0 \quad \forall i = 1, \dots, n$$

# Gradients

- The **grad** operator  $\nabla$  is the gradient operator in high dimensions

$$\nabla f(\mathbf{x}) = \begin{pmatrix} \frac{\partial f(\mathbf{x})}{\partial x_1} \\ \frac{\partial f(\mathbf{x})}{\partial x_2} \\ \vdots \\ \frac{\partial f(\mathbf{x})}{\partial x_n} \end{pmatrix}$$

- The partial derivatives (curly d's)

$$\frac{\partial f(\mathbf{x})}{\partial x_i}$$

means differentiate with respect to  $x_i$  treating all other components  $x_j$  as constants

# Least Squares Solution

- The least squared solution is give by

$$\begin{aligned}\nabla E(\boldsymbol{w}|\mathcal{D}) &= \nabla \|\boldsymbol{\epsilon}\|^2 = \nabla \|\boldsymbol{X}\boldsymbol{w} - \boldsymbol{y}\|^2 \\ &= \nabla (\boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} - 2\boldsymbol{w}^\top \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{y}^\top \boldsymbol{y}) \\ &= 2(\boldsymbol{X}^\top \boldsymbol{X} \boldsymbol{w} - \boldsymbol{X}^\top \boldsymbol{y}) = 0\end{aligned}$$

- Or

$$\boldsymbol{w} = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top \boldsymbol{y} = \boldsymbol{X}^+ \boldsymbol{y}$$

- $\boldsymbol{X}^+ = (\boldsymbol{X}^\top \boldsymbol{X})^{-1} \boldsymbol{X}^\top$  is known as the pseudo inverse

# Missing Bits of the Mathematics

- Note that  $\|a\|^2 = a^\top a = \sum_i a_i^2$

$$\begin{aligned}\|Xw - y\|^2 &= (Xw - y)^\top (Xw - y) = (w^\top X^\top - y^\top)(Xw - y) \\ &= w^\top X^\top Xw - 2w^\top X^\top y + y^\top y\end{aligned}$$

- Where we have used  $w^\top X^\top y = y^\top Xw$ ,  $\sum_{i,j} w_i X_{ji} y_j = \sum_{i,j} y_i X_{ij} w_j$
- Also  $\nabla w^\top M w = M w + M^\top w$
- If  $M = M^\top$  (i.e.  $M$  is symmetric) then  $\nabla w^\top M w = 2M w$
- As  $(AB)^\top = B^\top A^\top$  then  $(X^\top X)^\top = X^\top X^{\top\top} = X^\top X$  and  $X^\top X$  is symmetric

# Computing Gradients

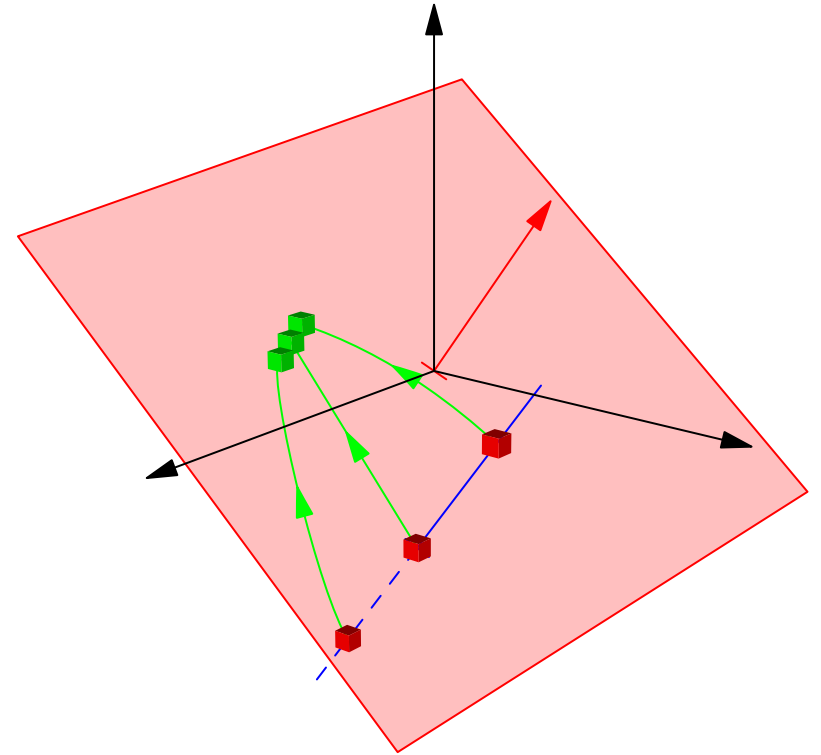
- To understand gradients we sometimes need to go back to components

$$\begin{aligned}\nabla \mathbf{w}^\top \mathbf{M} \mathbf{w} &= \begin{pmatrix} \frac{\partial}{\partial w_1} \\ \frac{\partial}{\partial w_2} \\ \frac{\partial}{\partial w_3} \\ \vdots \end{pmatrix} \sum_{i,j} w_i M_{ij} w_j = \begin{pmatrix} \sum_j M_{1j} w_j + \sum_i w_i M_{i1} \\ \sum_j M_{2j} w_j + \sum_i w_i M_{i2} \\ \sum_j M_{3j} w_j + \sum_i w_i M_{i3} \\ \vdots \end{pmatrix} \\ &= \mathbf{M} \mathbf{w} + \mathbf{M}^\top \mathbf{w}\end{aligned}$$

- It is tedious to compute these things component-wise, but when you need to understand what is going on then go back to the basics

# Outline

1. Mappings
2. **Linear Maps**

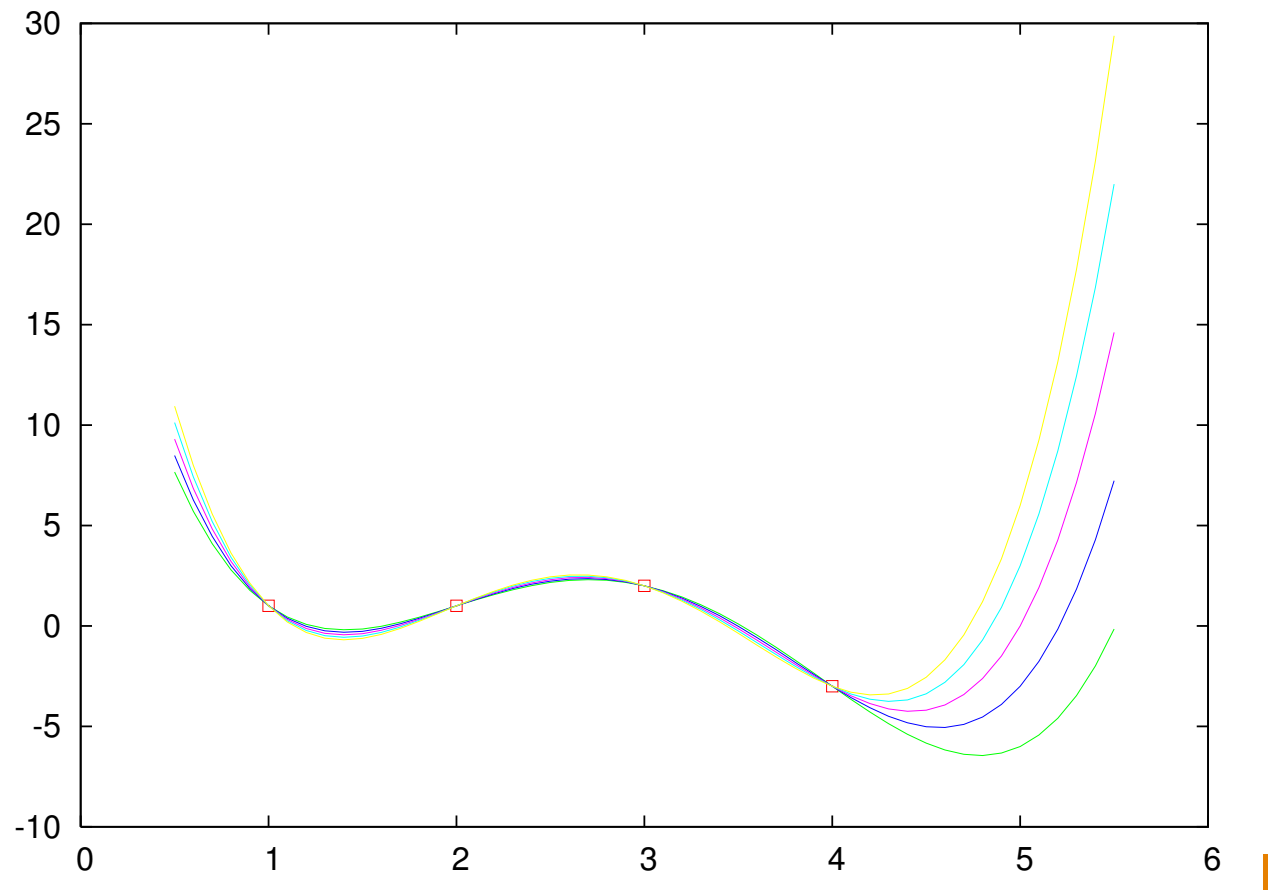


# Solving Inverse Problems

- Gauss showed us how to solve **over-constrained** problems (we have more observations than parameters)■
- We seek a solution which isn't necessarily exact but minimises an error■
- But, what if we have more parameters than observations■
- That is, we are **under-constrained**■
- Note that in some directions you might be over-constrained and in other directions under-constrained■
- This is very typical of most machine learning problems■

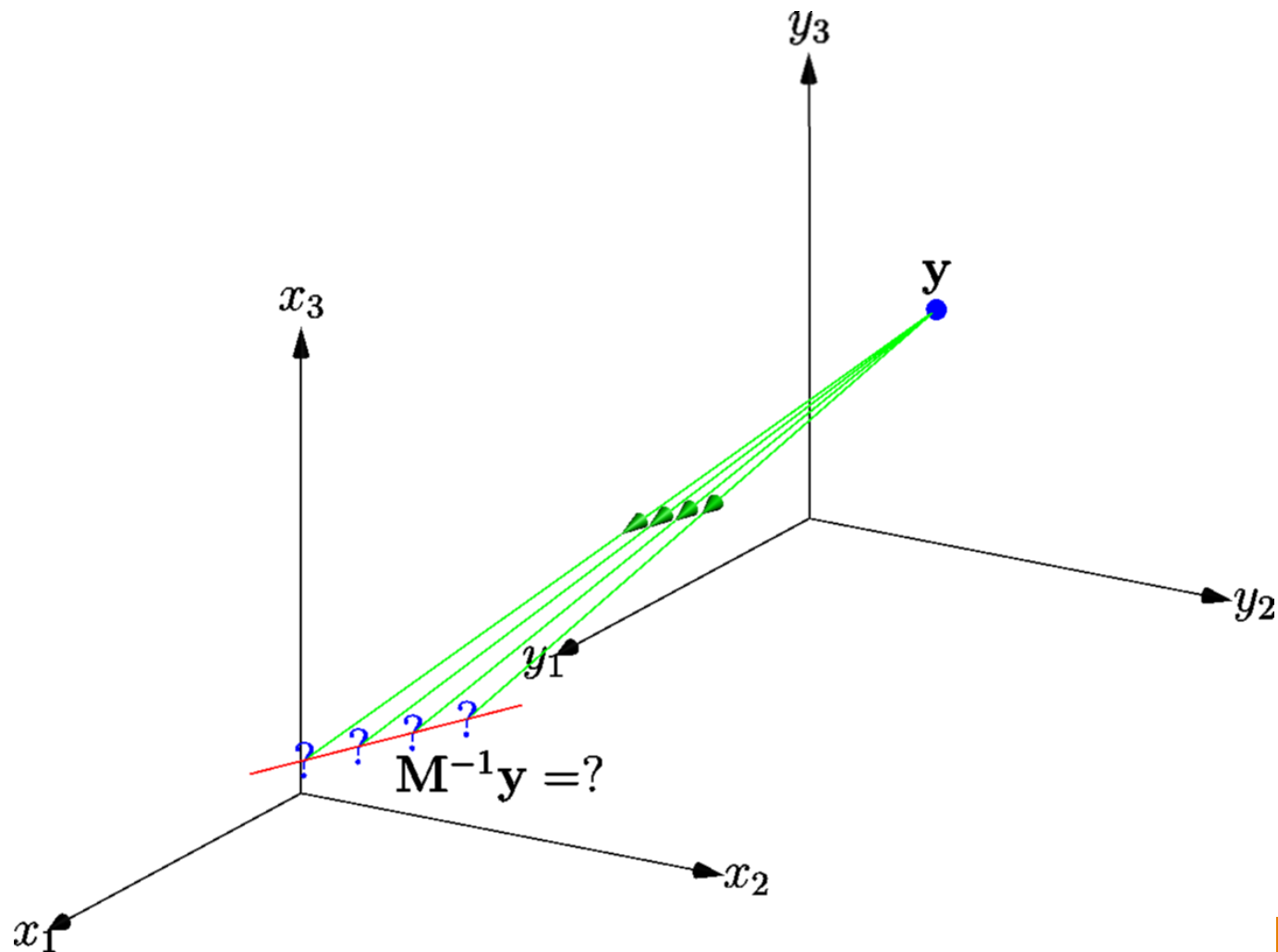
# Under Constrained Systems

- If we have less data-points than parameters then there will be multiple solutions



# What is the Inverse?

- Many points can map to the same points



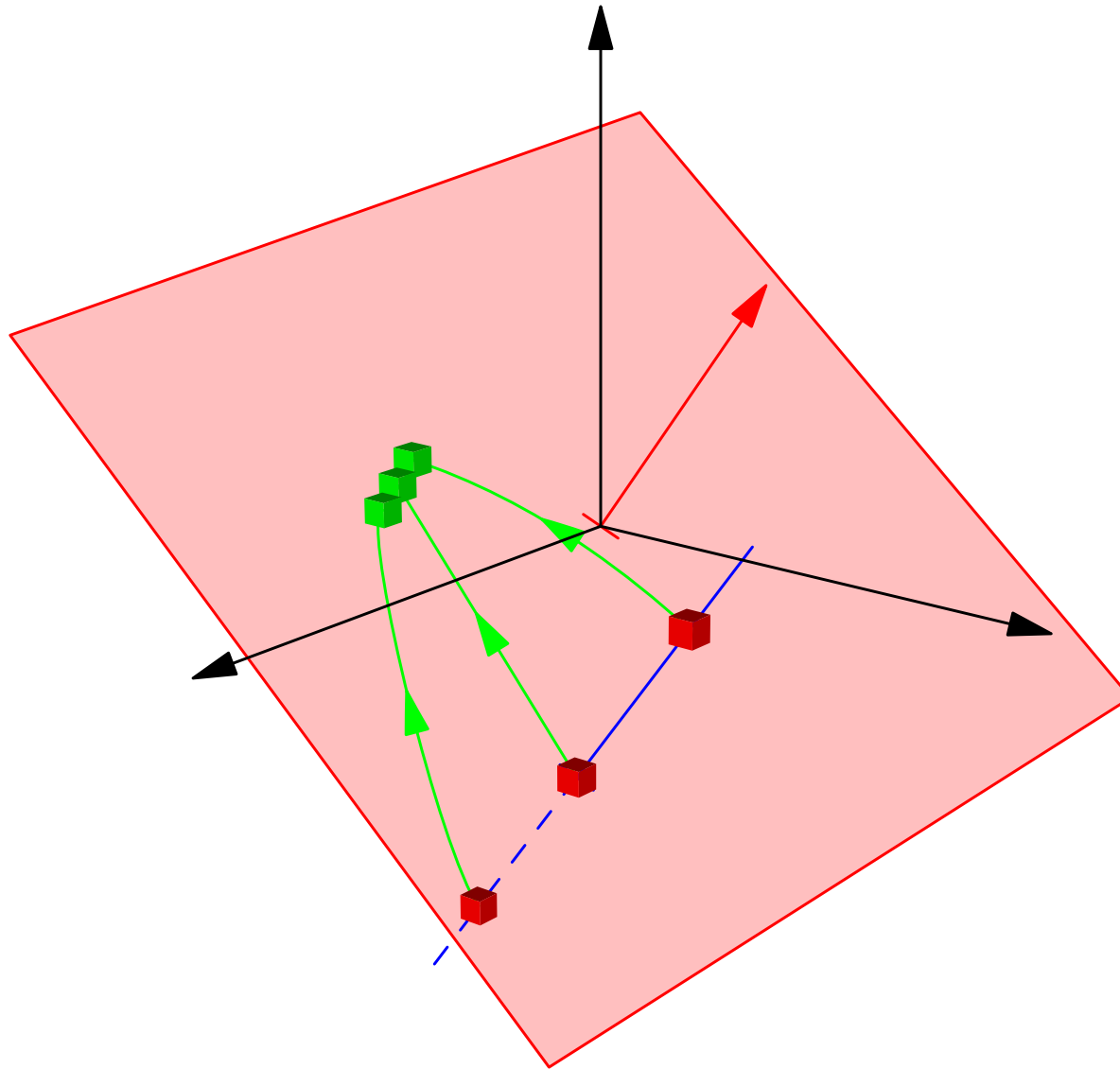
# Under-constrained Systems

- The system is **under-constrained**■
- We have more unknowns than equations■
- The inverse is not unique■
- Solving the inverse problem ( $w = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ ) is said to be **ill-posed**■
- The inverse  $(\mathbf{X}^T \mathbf{X})^{-1}$  doesn't exist■
- If we have a complicated learning machine and not sufficient data we often end with an ill-posed inverse problem (there are lots of sets of parameters that explain the data)■

# Ill-Conditions

- Singular matrices are rare (although they occur when we don't have enough data), but matrices that are close to being singular are common■
- If a matrix is close to singular it is ill-conditioned■
- Ill-conditioned matrices have some small eigenvalues■
- All points get contracted towards a plane■
- Large matrices are very often ill conditioned■

# III-Conditioned Matrices



# III-Conditioning in ML

- Ill-conditioning in machine learning occurs when a very small change in the learning data causes a large change in the predictions of the learning machine■
- In linear regression the matrix  $\mathbf{X}^T\mathbf{X}$  is ill-conditioned when we have as many data points as parameters■
- Much of machine learning is concerned with making learning machines better conditioned■
- Adding regularisers is one approach to achieve this■

# Summary

- Linear mappings are commonly used in machine learning algorithms such as regression■
- We will often meet the pseudo-inverse involving inverting  $X^T X$ ■
- They can be inherently unstable to noise in the inputs■